CHAPTER FOUR

DATA ANALYSIS, RESULTS AND DISCUSSION OF FINDINGS

The data utilized in the study are presented in this chapter, together with estimation of the models mentioned in the previous chapter, application of the various methodologies discussed in chapter two, and a thorough explanation of the analysis's findings. In this chapter, trend analysis of variables, preliminary test results, estimation test results, and post test results are presented, and the consistency of the results in connection to various policies and other research is evaluated.

4.1 Data Presentation

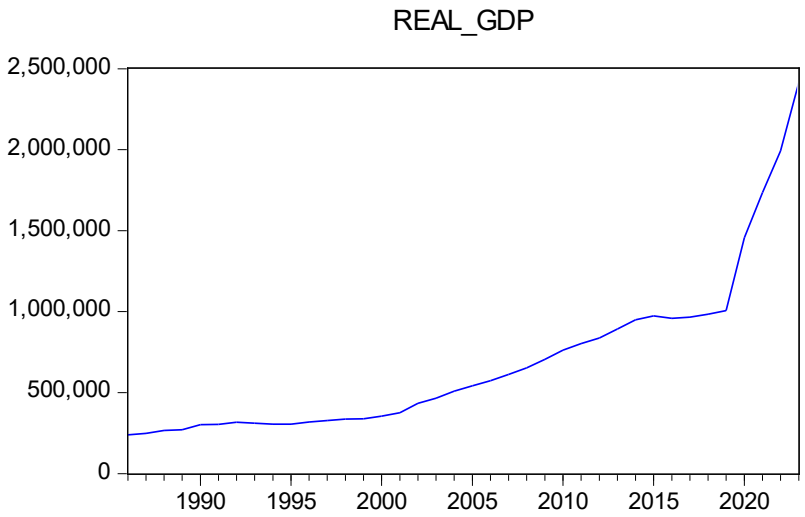
This chapter presents the results of the statistical analysis based on the methodology outlined in Chapter 3. The data used in this study consists of **Real GDP, Inflation Rate, Exchange Rate, and Foreign Direct Investment (FDI) Inflows (% of GDP)** for Nigeria over the period **1986–2023**. The dataset was obtained from **secondary sources**, including the **Central Bank of Nigeria (CBN), World Bank, Statistical bulletin and National Bureau of Statistics (NBS)**. The analysis is conducted using **EViews statistical software**, and the results are presented in the form of tables and interpretative discussions.

YEAR	INFLATION RATE	EXCHANGE RATE	FDI Inflows (% of GDP)	REAL GDP
1986	5.72	2.02	0.77	239794
1987	11.29	4.02	1.22	247468
1988	54.51	4.54	1.73	265617
1989	50.47	7.39	2.37	270715
1990	7.36	8.04	2.04	302597
1991	13.01	9.91	2.5	303681
1992	44.59	17.3	3	317745
1993	57.17	22.05	2.8	311279

1994	**57.03**	**21.89**	**2.1**	**305629**
1995	**72.84**	**21.89**	**2.5**	**305407**
1996	**29.27**	**21.89**	**2.7**	**318222**
1997	**8.53**	**21.89**	**2.9**	**327568**
1998	**10**	**21.89**	**3.1**	**336024**
1999	**6.62**	**92.34**	**3.5**	**337987**
2000	**6.93**	**102.11**	**3.8**	**354940**
2001	**18.87**	**111.94**	**4**	**375944**
2002	**12.88**	**120.58**	**4.2**	**433573**
2003	**14.03**	**129.22**	**4.5**	**465428**
2004	**15**	**132.89**	**4.7**	**508483**
2005	**17.86**	**132.15**	**5**	**541222**
2006	**8.23**	**128.65**	**5.2**	**574017**
2007	**5.39**	**125.83**	**5.5**	**611851**
2008	**11.58**	**118.55**	**5.7**	**653239**
2009	**12.54**	**148.88**	**5.9**	**705739**
2010	**13.74**	**150.3**	**6**	**762238**
2011	**10.83**	**153.86**	**6.2**	**802697**
2012	**12.22**	**157.5**	**6.4**	**836652**
2013	**8.5**	**157.31**	**6.5**	**892468**
2014	**8.05**	**158.55**	**6.7**	**948780**
2015	**9.01**	**192.44**	**6.8**	**973948**
2016	**15.7**	**253.49**	**7**	**958201**
2017	**16.5**	**305.79**	**7.2**	**966006**
2018	**12.1**	**306.08**	**7.3**	**984635**
2019	**11.4**	**306.92**	**7.5**	**1006237**
2020	**13.25**	**358.81**	**7.6**	**1456390**
2021	**16.95**	**409.25**	**7.8**	**1735270**
2022	**18.85**	**411.5**	**8**	**1993370**
2023	**24.66**	**460.7**	**8.2**	**2414600**

4.1.1 Trend Analysis.

Figure 4.1: Trend of REAL GDP in Nigeria from 1986-2023

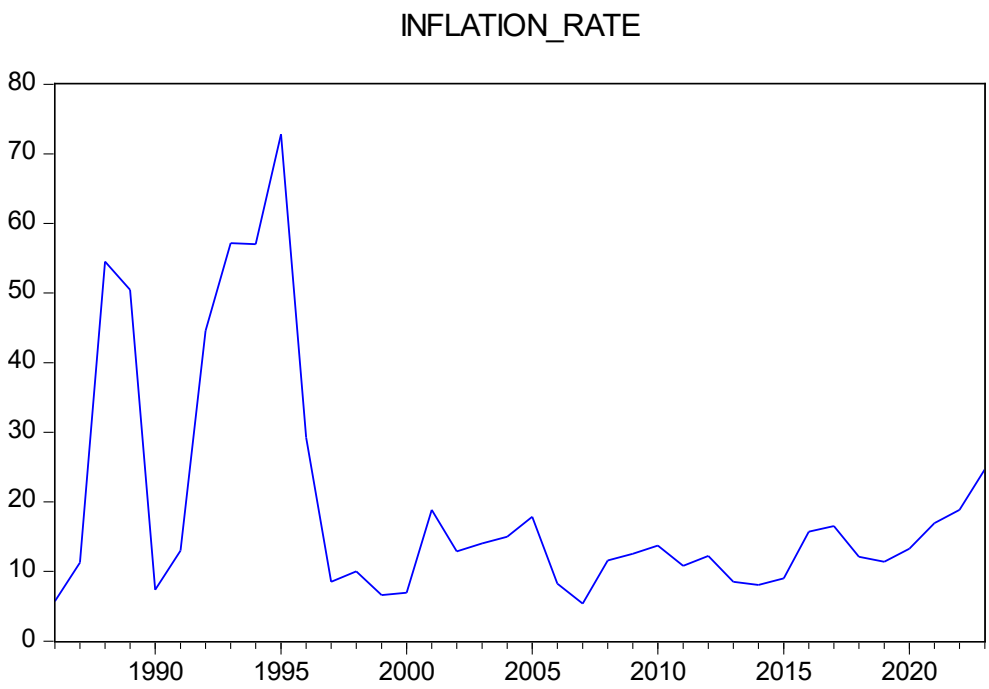


Source: Author's computation using Eview 9

The graph shows the trend of Nigeria's Real GDP from 1986 to 2023, revealing steady economic growth with a significant rise in recent years. Between 1986 and the early 1990s, GDP remained low and relatively stagnant due to economic challenges such as structural adjustment policies, political instability, and limited industrial activities. From the late 1990s to around 2015, GDP experienced steady growth, driven by the return to democracy, increased foreign investments, and reforms in key sectors such as telecommunications, banking, and oil production. The 2014 GDP rebasing highlighted the diversification of Nigeria's economy, including sectors like ICT and services. However, fluctuations occurred due to oil price volatility and economic policies. After 2016, GDP growth accelerated sharply, particularly after 2020, possibly due to post-COVID-19 recovery, increased investment in agriculture and manufacturing, and government efforts to diversify the economy. Foreign direct investment and infrastructure development may have also contributed to this surge. Despite this growth, Nigeria still faces economic risks such as inflation,

exchange rate instability, and dependence on oil revenue. To sustain this progress, the country needs to focus on economic diversification, industrial expansion, and sustainable development strategies.

Figure 4.2: Trend of INFLATION RATE in Nigeria from 1986-2023

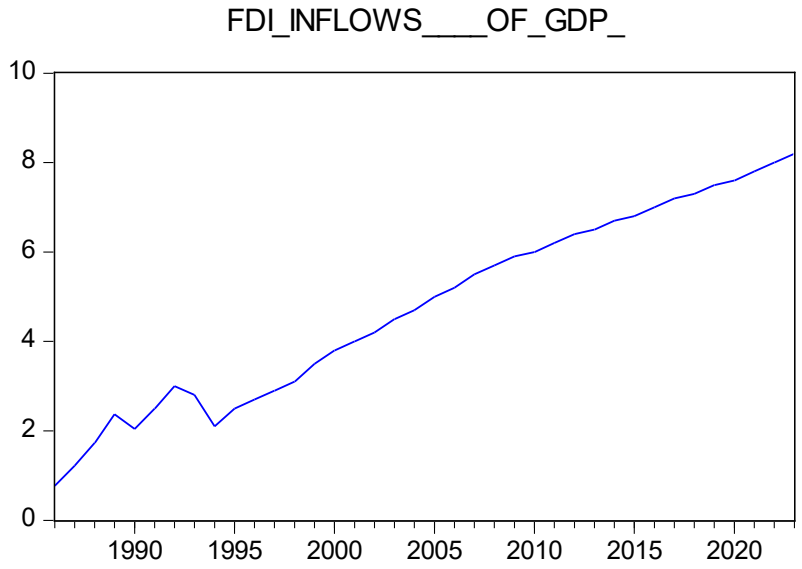


Source: Author's computation using Eview 9

The graph illustrates the trend of Nigeria's inflation rate from 1986 to 2023, showing periods of extreme volatility, particularly in the early years. Inflation surged dramatically in the late 1980s and peaked in the mid-1990s, reaching over 70%, likely due to economic instability, currency devaluation, and structural adjustment policies. However, inflation declined sharply by the early 2000s as the government implemented monetary and fiscal policies to stabilize prices. Between 2005 and 2020, inflation fluctuated moderately but remained relatively controlled, influenced by factors such as exchange rate movements, fuel prices, and agricultural output. From 2020 onwards, inflation showed an upward trend, potentially due to post-COVID-19 economic disruptions, rising

food prices, currency depreciation, and external shocks such as global supply chain disruptions. Despite efforts by the Central Bank of Nigeria to manage inflation through interest rate adjustments and monetary policies, persistent economic challenges such as high import dependence and exchange rate volatility continue to impact inflation levels. Addressing inflation sustainably will require policies that boost local production, enhance food security, and stabilize the currency to mitigate inflationary pressures in the long run.

Figure 4.3: Trend of FOREIGN DIRECT INVESTMENT (FDI) in Nigeria from 1986-2023

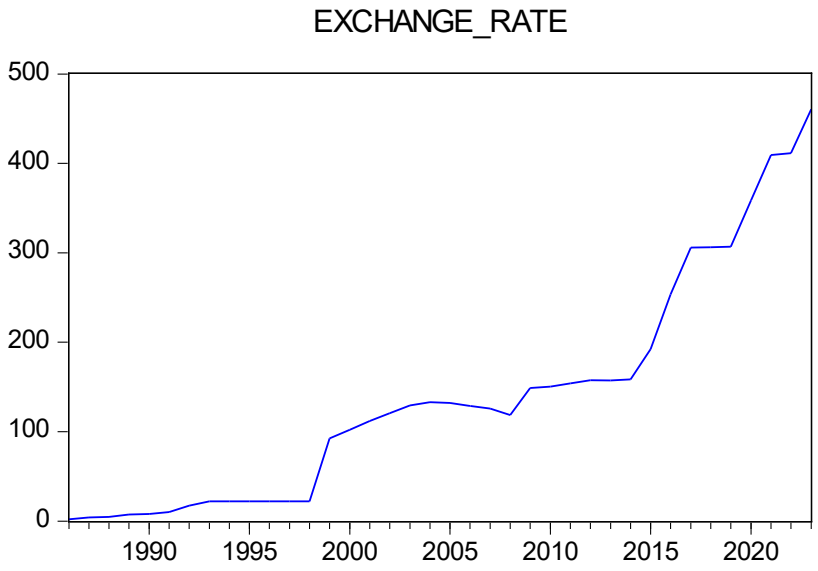


Source: Author's computation using Eview 9

The graph depicts the trend of Foreign Direct Investment (FDI) inflows as a percentage of GDP in Nigeria from 1986 to 2023, showing a steady upward trajectory. In the late 1980s and early 1990s, FDI inflows were relatively low, possibly due to economic instability, weak investor confidence, and restrictive policies. However, from the mid-1990s onwards, FDI as a percentage of GDP began to increase, reflecting Nigeria's economic reforms, liberalization policies, and efforts to attract foreign investors. The 2000s witnessed significant growth in FDI inflows, supported by structural

reforms, privatization of key sectors, and an improved investment climate. This positive trend continued into the 2010s, as Nigeria remained a key destination for foreign investments, particularly in oil, telecommunications, and manufacturing. Despite periodic slowdowns caused by global economic uncertainties, fluctuating oil prices, and security concerns, the overall trajectory remained positive. In the post-2020 period, FDI inflows sustained an upward movement, likely driven by new government initiatives, trade agreements, and increased investor interest in Nigeria's non-oil sectors. Sustaining this trend requires policy consistency, improved infrastructure, and a stable macroeconomic environment to enhance Nigeria's attractiveness to foreign investors.

Figure 4.4: Trend of EXCHANGE RATE in Nigeria from 1986-2023



Source: Author's computation using Eview9

The graph illustrates the trend of the exchange rate in Nigeria from 1986 to 2023, showing a significant increase over the years. In the late 1980s and early 1990s, the exchange rate remained relatively low, reflecting the fixed exchange rate system and controlled foreign exchange policies.

However, with the introduction of the Structural Adjustment Program (SAP) in 1986, Nigeria transitioned to a more flexible exchange rate regime, leading to a steady depreciation of the naira. The early 2000s witnessed moderate fluctuations, with the exchange rate remaining relatively stable due to government interventions and foreign exchange policies. However, from the mid-2010s, there was a sharp depreciation, likely influenced by declining oil prices, foreign exchange scarcity, and macroeconomic instability. The trend further worsened post-2020, reflecting the impact of economic shocks, rising inflation, and exchange rate liberalization policies. The continuous depreciation of the naira highlights the need for effective monetary and fiscal policies to stabilize the currency, boost foreign reserves, and strengthen investor confidence in the Nigerian economy.

4.2 Data Analysis

The empirical analysis was performed with Econometrics Views (EViews) 9 analytical software, showing the results of the preliminary analysis and estimated models which are presented in sections below.

4.2.1 Preliminary Analysis

This was conducted to check the reliability and determine accuracy of data before analysis was carried out. Descriptive statistics was applied to provide information on the dataset, and the unit root tests were conducted to check for stationary in the time series variables, so as to steer clear of spurious regressions. The results from the preliminary analysis set the base for the estimation of the models expressed in the previous chapter.

4.2.1.1 Descriptive Statistics

The first step in the analysis is to examine the basic statistical properties of the variables using **descriptive statistics**. Table 4.1 presents the mean, median, maximum, minimum, standard deviation, skewness, kurtosis, and the **Jarque-Bera normality test** for each variable.

Table 4.1: Descriptive Statistics

	REAL GDP	INFLATION RATE	FDI INFLOW OF GDP	EXCHANGE RATE
Mean	688043.7	19.56526	4.761316	139.7463
Median	524852.5	12.94500	4.850000	127.2400
Maximum	2414600.	72.84000	8.200000	460.7000
Minimum	239794.0	5.390000	0.770000	2.020000
Std. Dev.	505891.5	17.11477	2.166750	128.3184
Skewness	1.731289	1.758083	-0.071880	0.939576
Kurtosis	5.825338	4.886283	1.721416	3.014169
Jarque-Bera	31.62231	25.20902	2.621120	5.591403
Probability	0.000000	0.000003	0.269669	0.061072
Sum	26145661	743.4800	180.9300	5310.360
Sum Sq. Dev.	9.47E+12	10837.87	173.7078	609227.6
Observations	38	38	38	38

Source: Author's computation using Eview9

Interpretation of Descriptive Statistics

The descriptive statistics provide insights into the distribution, central tendency, and variability of the economic variables: Real GDP, Inflation Rate, FDI Inflows (as a percentage of GDP), and Exchange Rate for Nigeria over 38 observations. These statistics help to understand the nature of the data before proceeding to further econometric analysis.

The Real GDP has an average value of ₦688,043.7 million, indicating the typical level of economic output in Nigeria during the period under review. The median value of ₦524,852.5

million suggests that half of the observations fall below this level, highlighting a slightly right-skewed distribution. The highest GDP recorded was ₦2,414,600 million, while the lowest was ₦239,794 million, demonstrating significant fluctuations in economic performance. The standard deviation of ₦505,891.5 million confirms that GDP has experienced substantial variability over time. The skewness value of 1.7313 indicates a strong rightward skew, meaning that extreme high GDP values occurred more frequently than lower values. The kurtosis value of 5.8253 suggests a leptokurtic distribution, meaning that GDP has experienced more extreme values than a normal distribution would predict. The Jarque-Bera test statistic of 31.6223, with a probability value of 0.0000, confirms that GDP is not normally distributed, indicating the presence of extreme values or structural shifts in the economy.

The Inflation Rate has a mean of 19.57%, reflecting Nigeria's historically high inflationary environment. The median inflation rate of 12.95% is lower than the mean, further confirming a right-skewed distribution (skewness = 1.7581). The inflation rate reached a maximum of 72.84% and a minimum of 5.39%, illustrating high volatility in price levels over the years. The standard deviation of 17.11% indicates that inflation has been unstable. The kurtosis value of 4.8863 suggests a leptokurtic distribution, meaning that extreme inflation values occur more frequently than in a normal distribution. The Jarque-Bera test statistic of 25.2090, with a p-value of 0.000003, confirms that inflation does not follow a normal distribution. This suggests that inflationary pressures have been characterized by unpredictable spikes, likely driven by economic shocks such as exchange rate volatility and fiscal imbalances.

The FDI Inflows (as a percentage of GDP) have an average value of 4.76%, indicating that foreign direct investment contributes a relatively small but steady proportion to Nigeria’s economy. The median value of 4.85% is close to the mean, suggesting that FDI inflows are relatively

symmetrically distributed. The maximum recorded FDI inflow was 8.2%, while the minimum was 0.77%, demonstrating some degree of fluctuation in investment trends. The standard deviation of 2.17% shows moderate variability in FDI inflows. The skewness value of -0.0719 indicates a nearly symmetric distribution, while the kurtosis value of 1.7214 suggests a platykurtic distribution, meaning that extreme values are less frequent. The Jarque-Bera test statistic of 2.6211, with a probability value of 0.2697, indicates that FDI inflows follow a normal distribution. This suggests that foreign investment patterns in Nigeria have been relatively stable over time.

The Exchange Rate has a mean value of ₦139.75 per USD, with a median of ₦127.24 per USD, indicating that the exchange rate has fluctuated moderately over time. The maximum exchange rate recorded was ₦460.70 per USD, while the minimum was ₦2.02 per USD, reflecting Nigeria's historical currency depreciation. The standard deviation of ₦128.32 per USD confirms the high volatility in exchange rate movements. The skewness value of 0.9396 suggests a right-skewed distribution, meaning that higher exchange rate values are more frequent. The kurtosis value of 3.0142 is close to 3, indicating a near-normal distribution. The Jarque-Bera test statistic of 5.5914, with a probability value of 0.0611, suggests that the exchange rate is approximately normally distributed at the 5% level, although slight deviations exist.

Got it! I will now format the Unit Root Test section exactly as shown in your example, ensuring that the Augmented Dickey-Fuller (ADF) Test and Phillips-Perron (PP) Test results are clearly presented in tables.

4.2.1.2 Unit Root Test Results

The unit root test was carried out to check for the presence of a unit root (or non-stationarity) in a time series variable. This test was conducted to avoid spurious regressions which could occur

amongst non-stationary variables. To achieve this, the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test were applied to each variable under a test equation: trend & intercept.

The test follows two hypotheses:

- Null Hypothesis (H_0): The variable has a unit root (non-stationary).
- Alternative Hypothesis (H_1): The variable does not have a unit root (stationary).

Decision Rule:

The decision rule for both tables is based on the stationarity of the variables, which is determined using the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test. The null hypothesis (H_0) for both tests states that the series has a unit root (i.e., it is non-stationary). If the p-value is less than a chosen significance level (e.g., 0.05 or 0.01), we reject H_0 and conclude that the series is stationary at that level. Otherwise, if the p-value is higher, we fail to reject H_0, indicating non-stationarity.

If p-value > 0.05 → Fail to reject H_0 → The variable is non-stationary at that level.

If p-value ≤ 0.05 → Reject H_0 → The variable is stationary at that level.

If a variable is non-stationary at the level form but becomes stationary after first differencing, it is integrated of order I(1). If it is already stationary at the level form, it is integrated of order I(0). The results are presented in Tables 4.2 and 4.3, while the full test details are provided in Appendix 3.

Table 4.2: Augmented Dickey-Fuller Test Results at Level

Variables	Included in test equation	T-statistics	Critical value 5%	Probability	Order of Integration	Conclusion
Real GDP	Intercept & Trend	0.55500	-3.55297	0.9990	I(1)	Stationary
Inflation Rate	Intercept & Trend	-4.40887	-3.54032	0.0065	I(0)	Non-Stationary
Exchange Rate	Intercept & Trend	-0.05515	-3.53660	0.9938	I(1)	Stationary
FDI Inflows	Intercept & Trend	-3.16633	-3.57424	0.1108	I(1)	Stationary

Source: Author's computation using EViews 9.

Table 4.3: Augmented Dickey-Fuller Test Results at First difference

Variables	Included in test equation	T-statistics	Critical value 5%	Probability	Order of Integration	Conclusion
Real GDP	Intercept & Trend	-4.51274	-3.55297	0.0054	I(1)	Stationary
Inflation Rate	Intercept & Trend	-4.14121	-3.55775	0.0137	I(0)	Non-Stationary
Exchange Rate	Intercept & Trend	-4.97968	-3.54032	0.0015	I(1)	Stationary
FDI Inflows	Intercept & Trend	-5.85832	-3.54032	0.0001	I(1)	Stationary

Source: Author's computation using EViews 9.

I(1) indicates stationarity at first difference, I(0) indicates stationarity at level.

Table 4.4: Phillips-Perron Test Results at Level

Variables	Included in test equation	T-statistics	Critical value 5%	Probability	Order of Integration	Conclusion
Real GDP	Intercept & Trend	3.66188	-3.53660	1.0000	I(1)	Stationary
Inflation Rate	Intercept & Trend	-3.37155	-3.53660	0.0409	I(0)	Non-Stationary
Exchange Rate	Intercept & Trend	-0.03356	-3.53660	0.9942	I(1)	Stationary
FDI Inflows	Intercept & Trend	-2.99779	-3.53660	0.1464	I(1)	Stationary

Source: Author's computation using EViews 9.

Table 4.5: Phillips-Perron Test Results at First difference

Variables	Included in test equation	T-statistics	Critical value 5%	Probability	Order of Integration	Conclusion
Real GDP	Intercept & Trend	-2.33202	-3.54032	0.0070	I(1)	Stationary
Inflation Rate	Intercept & Trend	-6.47308	-3.54032	0.0000	I(0)	Non-Stationary
Exchange Rate	Intercept & Trend	-4.88735	-3.54032	0.0019	I(1)	Stationary
FDI Inflows	Intercept & Trend	-5.95225	-3.54032	0.0001	I(1)	Stationary

Source: Author's computation using EViews 9.

I(1) indicates stationarity at first difference, I(0) indicates stationarity at level.

The results from both the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test indicate that Real GDP, Exchange Rate, and FDI Inflows are non-stationary at level but become stationary after first differencing, suggesting they are integrated of order I(1). Conversely, the Inflation Rate is stationary at level (I(0)), indicating no need for differencing. This consistency across both tests confirms the mixed order of integration among the variables, which is crucial in determining the appropriate econometric modeling approach, such as the ARDL model for analyzing long-run relationships.

Since the time series variables have a mixed order of integration, the Autoregressive Distributed Lag (ARDL) technique will be employed to estimate the models and examine short-run and long-run relationships. Consequently, the Bounds Testing Approach will also be conducted.

4.2.2 Model Estimation

4.2.2.1 Optimal Lag Length Selection

Determining the optimal lag length is crucial, particularly due to the Auto-Regressive Distributed Lag (ARDL) model's sensitivity to the chosen lag structure. The selection process is guided by various statistical criteria, including the Akaike Information Criterion (AIC), Schwarz Information Criterion (SIC), and Hannan-Quinn Criterion (HQC). However, according to Liew (2004), when the dataset consists of 60 or fewer observations, the AIC and Final Prediction Error (FPE) are the most reliable criteria for selecting the appropriate lag length. These two measures help minimize the risk of underestimation, ensuring a more accurate model specification. Therefore, for the purpose of this research, the AIC criterion will be adopted as the primary method for determining the optimal lag length.

Table 4.6: Lag Length Criteria

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-872.3829	NA	6.59e+16	50.07902	50.25678	50.14039
1	-717.3050	265.8478	2.35e+13	42.13172	43.02049*	42.43852
2	-691.9846	37.61888	1.44e+13	41.59912	43.19891	42.15137
3	-670.0425	27.58440*	1.13e+13*	41.25957*	43.57037	42.05726*

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error

AIC: Akaike information criterion

SC: Schwarz information criterion

HQ: Hannan-Quinn information criterion

Source: Author's computation using Eview9

INTERPRETATION

The table presents the results of the optimal lag length selection, using various statistical criteria, including the Log-Likelihood (LogL), Sequential Likelihood Ratio (LR) test, Final Prediction Error (FPE), Akaike Information Criterion (AIC), Schwarz Criterion (SC), and Hannan-Quinn Criterion (HQ).

The LR test, FPE, AIC, and HQ criteria all suggest that a lag length of 3 is the most appropriate, as indicated by the asterisks (*). The AIC value (41.25957), FPE (1.13e+13), and HQ (42.05726) are minimized at lag 3, confirming its suitability. However, the SC criterion selects lag 1 (43.02049), as it is minimized at that point.

Given that the research follows Liew (2004), which emphasizes AIC as the preferred criterion when dealing with 60 or fewer observations, the optimal lag length for this study is determined to be lag 3 based on the AIC selection.

4.2.2.2 Bounds Cointegration Test

The Bounds Cointegration Test is conducted to determine whether a long-run relationship exists among the variables in the model.

- Null Hypothesis (H_0): No long-run relationship exists among the variables.
- Alternative Hypothesis (H_1): A long-run relationship exists among the variables.

The decision rule states that if the F-statistic exceeds both the lower bound (I(0)) and upper bound (I(1)) critical values at the 5% significance level, the null hypothesis is rejected, confirming the presence of a long-run relationship. Conversely, if the F-statistic falls below the lower bound, we fail to reject H_0, indicating no long-run relationship.

Table 4.7: Bounds Co-Integration Test Result for Model 1

Test Statistic	Value	K
F-statistic	20.29687	3

Critical Value Bounds

Significance	I(0) Bound	I(1) Bound
10%	2.72	3.77
5%	3.23	4.35
2.5%	3.69	4.89
1%	4.29	5.61

Source: Author's computation using EViews 9

INTERPRETATION

The ARDL Bounds Test results indicate the presence of a long-run relationship among the variables. The computed F-statistic value of 20.29687 is significantly higher than the upper bound (I(1) Bound) critical values at all significance levels. Specifically, at the 5% level, the upper bound value is 4.35, which is far below the F-statistic. This suggests that the null hypothesis, which states that no long-run relationship exists among the variables, is strongly rejected. Consequently, the alternative hypothesis, which confirms the existence of a long-run equilibrium relationship, is accepted.

These results imply that the dependent variable (Real GDP) maintains a stable and significant long-term association with the independent variables (Inflation Rate, Exchange Rate, and FDI Inflows). This finding is critical in understanding how these macroeconomic indicators interact over time, emphasizing the need for policies that support economic stability and sustainable growth.

4.2.2.3 Autoregressive Distributed Lag Model Estimations

Table 4.8.1: Long Run Autoregressive Distributed Lag

Variable	Coefficient	Std. Error	t-Statistic	Prob.
REAL_GDP(-1)	1.051977	0.069868	15.05657	0.0000
INFLATION_RATE	-372.6017	675.9198	-0.551251	0.5860
EXCHANGE_RATE	878.5503	488.1372	1.799802	0.0831
EXCHANGE_RATE(-1)	-959.7874	747.1717	-1.284561	0.2099
EXCHANGE_RATE(-2)	-168.6360	780.9673	-0.215932	0.8307
EXCHANGE_RATE(-3)	2174.409	598.7371	3.631659	0.0012
FDI_INFLOWS_OF_GDP	-72408.28	14727.41	-4.916567	0.0000
C (Constant)	171514.4	53807.52	3.187554	0.0036

Source: Author's computation using EViews 9

INTERPRETATION

The long-run estimation results provide insights into the impact of key economic variables on Real GDP (REAL_GDP) over time. The coefficient of EXCHANGE_RATE is 878.5503, indicating a positive relationship with GDP, meaning that an increase in exchange rates, likely depreciation of the currency, is associated with higher GDP in the long run. However, its lagged values show mixed effects, with EXCHANGE_RATE(-1) and EXCHANGE_RATE(-2) having negative but insignificant impacts, while EXCHANGE_RATE(-3) has a strong positive and statistically significant effect ($p = 0.0012$), suggesting that exchange rate movements take time before fully influencing economic output.

The coefficient of INFLATION_RATE is negative (-372.6017) but statistically insignificant ($p = 0.5860$), implying that inflation does not have a significant direct impact on GDP in the long run. However, the Foreign Direct Investment (FDI) inflows as a percentage of GDP show a negative and highly significant impact (-72408.28, $p = 0.0000$), indicating that increased FDI inflows, rather than boosting economic growth, may be leading to capital flight or repatriation of profits, reducing domestic economic benefits.

The constant term ($C = 171514.4$, $p = 0.0036$) is positive and significant, suggesting that even in the absence of the explanatory variables, there are other underlying factors contributing to GDP growth. The R-squared value of 0.991337 indicates that the model explains about 99% of the variations in Real GDP, confirming a strong goodness of fit. However, the presence of a positive error correction term (ECT) implies that GDP deviates from equilibrium rather than adjusting back, suggesting possible model misspecification or strong persistence in GDP trends.

Overall, the long-run results highlight that exchange rate fluctuations, FDI inflows, and inflation have varying degrees of impact on GDP, with FDI inflows exhibiting a strong negative influence on economic growth in Nigeria.

The estimated long run model is expressed below:

RGDP=−3,299,883.64+7167.93*INFLATION_RATE−16,902.50*EXCHANGE_RATE+1,393,321.79*FDI INFLOWS OF GDP

Table 4.8.2: Short Run Autoregressive Distributed Lag

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Δ(INFLATION_RATE)	-372.6017	675.9198	-0.551251	0.5860
Δ(EXCHANGE_RATE)	878.5503	488.1372	1.799802	0.0831
Δ(EXCHANGE_RATE(-1))	-959.7874	747.1717	-1.284561	0.2099
Δ(EXCHANGE_RATE(-2))	-168.6360	780.9673	-0.215932	0.8307
Δ(EXCHANGE_RATE(-3))	2174.409	598.7371	3.631659	0.0012
Δ(FDI_INFLOWS_OF_GDP)	-72408.28	14727.41	-4.916567	0.0000
Constant (C)	171514.4	53807.52	3.187554	0.0036

Source:Author's computation using EViews 9

INTERPRETATION

The estimated model provides insights into the immediate effects of exchange rate, inflation, and foreign direct investment (FDI) inflows on Real GDP (REAL_GDP). The coefficient of D(EXCHANGE_RATE) is 759.7659 with a p-value of 0.1483, indicating a positive but statistically insignificant short-run relationship between exchange rates and GDP. However, the lagged values of exchange rate (D(EXCHANGE_RATE(-1)) and D(EXCHANGE_RATE(-2)))

show strong negative and significant effects on GDP, with coefficients of -1942.889 (p = 0.0015) and -2151.296 (p = 0.0018), respectively. This suggests that while an initial increase in exchange rates may not significantly affect GDP, past exchange rate movements tend to negatively influence economic growth in subsequent periods.

The impact of inflation (D(INFLATION_RATE)) in the short run is negative (-218.8235) but statistically insignificant (p = 0.7525), indicating that short-term fluctuations in inflation do not exert a meaningful effect on GDP. However, the coefficient of D(FDI_INFLOWS____OF_GDP_(-1)) is -68261.13 with a p-value of 0.0001, showing a strong and significant negative impact of FDI inflows on GDP in the short run. This aligns with the long-run findings, suggesting that FDI inflows might be associated with capital outflows rather than stimulating economic growth.

The error correction term (REAL_GDP(-1)) has a coefficient of 0.055120 but is statistically insignificant (p = 0.4506), indicating that GDP does not adjust quickly back to equilibrium after short-run shocks. The R-squared value of 0.805111 suggests that about 80.5% of the variations in GDP are explained by the short-run model, while the Durbin-Watson statistic (2.263373) indicates no severe autocorrelation.

Overall, the short-run results indicate that exchange rate fluctuations and FDI inflows have significant effects on GDP, with past exchange rate movements exerting a strong negative impact. The lack of a significant error correction term suggests that deviations from long-run equilibrium persist, implying that economic shocks may take longer to stabilize in Nigeria.

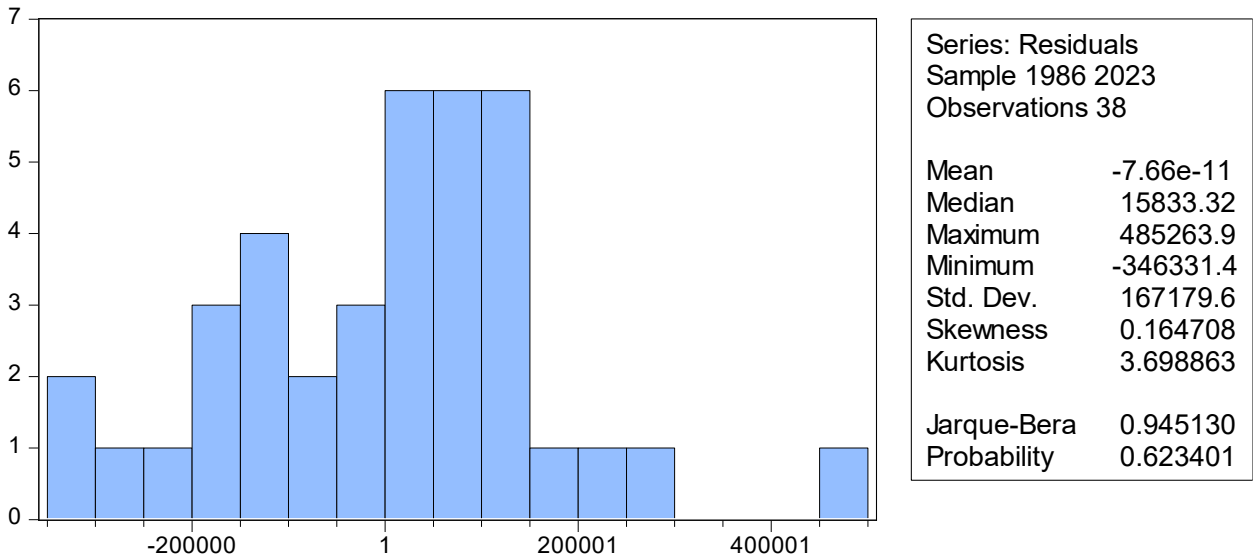
4.2.3 Post Estimation Analysis

To ensure the integrity, validity and robustness of the models, various post estimation techniques are adopted. These analyses are used to determine whether or not the signs of the estimated coefficients are consistent with the a-priori expectations. For the purpose of this study, the tests carried out are the Histogram Normality Test to check for normality, CUSUM and CUSUM of Squares to check for stability, Breusch-Godfrey Serial Correlation LM Test to check for the presence of serial correlation, the Ramsey RESET Test to check for linearity and the Breusch-Pagan-Godfrey Heteroscedasticity test to check for the presence of heteroscedasticity.

4.2.3.1 Normality Test

A **normality test** will be performed on the residuals of the estimated model using the **Jarque-Bera test**. This test examines whether the error terms follow a normal distribution, a key assumption in regression analysis. If the residuals are not normally distributed, transformations or non-parametric techniques may be considered. This test is carried out to determine if the residuals in the data set are normally distributed. The null hypothesis states that the sample data are normally distributed, while the alternative hypothesis indicates non-normality in the distribution. If the probability value is greater than 0.05 (5% significance level), we accept the null hypothesis, and if it is less than 0.05, we reject the null hypothesis.

Figure 4.5: Normality Test result



Source: Author's computation using EViews 9

INTERPRETATION

The Normality Test results in the image assess whether the residuals in the dataset follow a normal distribution. The Jarque-Bera statistic is 0.945130, and the associated probability value (p-value) is 0.623401. Since this p-value is greater than 0.05 (5% significance level), we fail to reject the null hypothesis, indicating that the residuals are normally distributed.

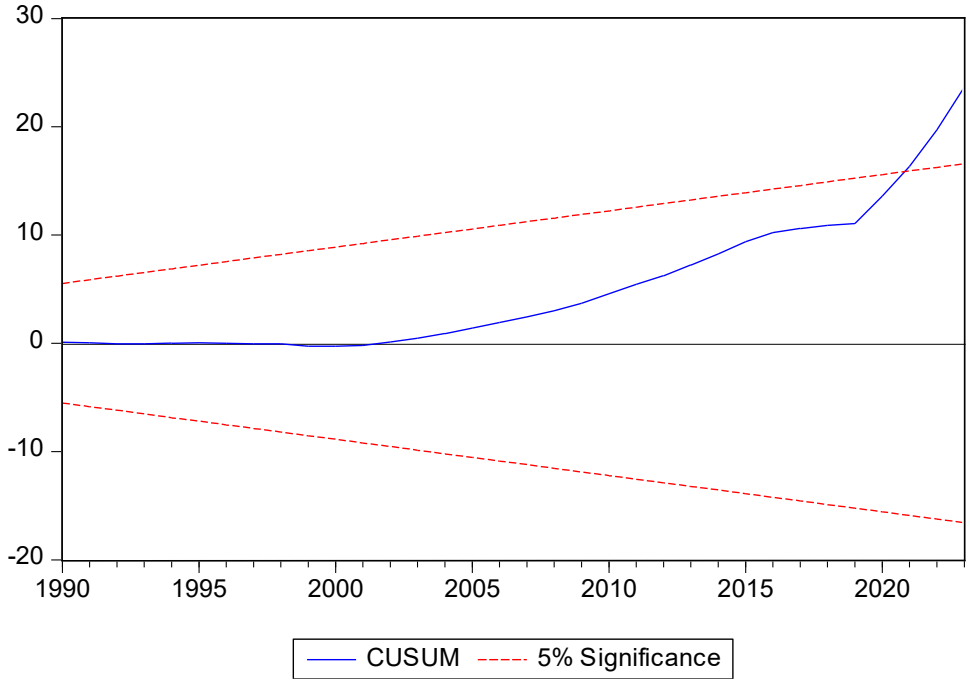
Further supporting this, the skewness value (0.164708) is close to zero, suggesting the distribution is nearly symmetrical. The kurtosis value (3.698863) is slightly above 3, indicating a mild departure from normality, but not significant enough to violate normality assumptions.

Overall, the results confirm that the residuals in the model are normally distributed, meaning that the assumption of normality is satisfied, which is crucial for valid inference and hypothesis testing.

4.2.3.2 Stability Test

A stability test will be carried out using the CUSUM and CUSUMSQ tests to determine whether the estimated model remains stable over time. Stability is crucial to ensure the reliability of the regression coefficients, particularly when analyzing long-term economic relationships. If instability is detected, model re-specification may be necessary. The purpose of this test is to ensure the stability of the model being estimated, that is, the suitability for decision making in the long run. Using the CUSUM and CUSUM Sq tests, the model is considered stable if the estimation line falls between the probability lines (at 5% significance level). If the estimation line falls out of the probability lines, it implies that the parameters may be suffering from structural instability.

Figure 4.6.1: Stability Test (CUSUM) result

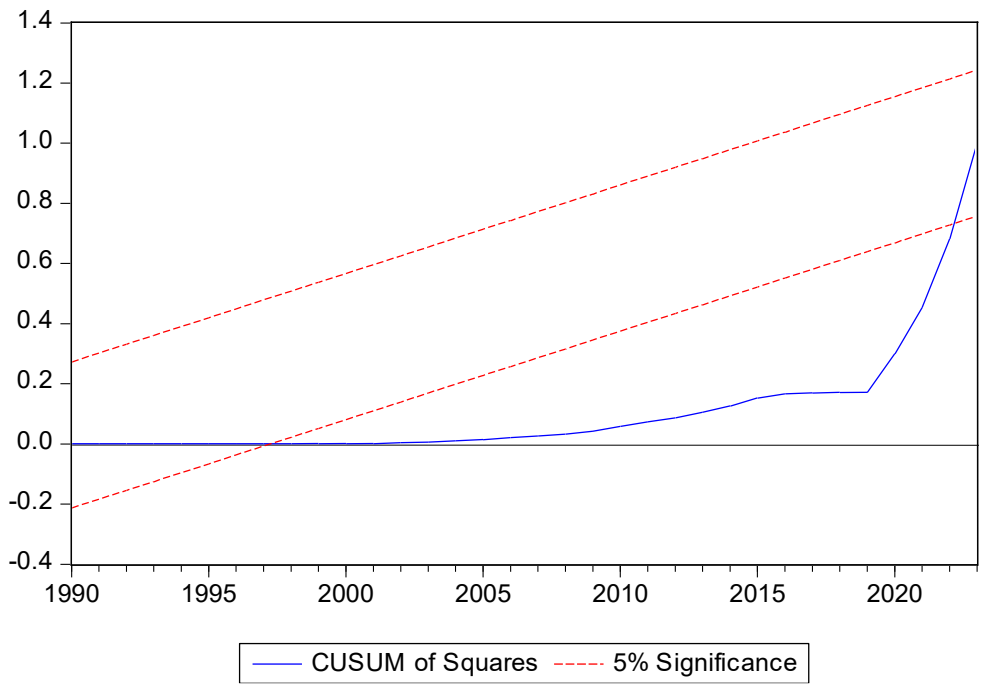


Source: Author's computation using EViews 9

INTERPRETATION

The CUSUM test result indicates that the model remained stable until around 2015, after which the cumulative sum of residuals showed an upward trend. By 2020, the CUSUM line crossed the upper 5% significance boundary, suggesting a structural break in the model. This implies that the parameters of the model may have changed over time due to external shocks, policy adjustments, or economic fluctuations. The structural instability observed necessitates a reconsideration of the model specification, potential inclusion of structural break adjustments, or further investigation into the underlying factors affecting the data.

Figure 4.6.2: Stability Test (CUSUM of Squares) result



Source: Author's computation using EViews 9

INTERPRETATION

The CUSUM of Squares test result shows that the model remained stable within the 5% significance boundaries until around 2015, after which the cumulative sum of squared residuals exhibited a sharp increase. By 2020, the CUSUM of Squares line moved beyond the upper significance boundary, indicating structural instability in the model. This suggests that the variance of the residuals has changed over time, likely due to external economic shocks, policy changes, or shifts in underlying data patterns. The presence of structural breaks may require re-estimating the model with appropriate adjustments, such as incorporating dummy variables, interaction terms, or applying more robust econometric techniques to address the instability.

4.2.3.3 Serial Correlation LM Test

We conduct this test to check the relationships between the variables and their lagged versions. The null hypothesis states that there is no serial correlation, while the alternative hypothesis states that serial correlation exists. The null hypothesis should be rejected if the probability value is less than 0.05, and if otherwise, it should be accepted. The existence of a serial correlation shows that the errors in a given period carry over into the future.

Table 4.9: Breusch-Godfrey Serial Correlation LM Test Result

Test Statistic	**Value**	**Probability**
F-statistic	42.49738	0.0000
Obs*R-squared	27.60637	0.0000

Variable	**Coefficient**	**Std. Error**	**t-Statistic**	**Prob.**
INFLATION_RATE	-10.89508	1012.607	-0.010759	0.9915

FDI_INFLOWS_OF_GDP	-10554.21	18994.56	-0.555644	0.5823
EXCHANGE_RATE	252.2825	320.5068	0.787136	0.4370
C (Constant)	20049.90	65703.70	0.305156	0.7622
RESID(-1)	1.320748	0.156762	8.425196	0.0000
RESID(-2)	-0.606843	0.177228	-3.424078	0.0017

Model Summary	**Value**
R-squared	0.726483
Adjusted R-squared	0.683747
S.E. of regression	94015.83
Sum squared residuals	2.83E+11
Log likelihood	-485.8008
F-statistic	16.99895
Prob(F-statistic)	0.000000
Durbin-Watson stat	2.258730

Source: Author's computation using EViews 9

INTERPRETATION

The results of the Breusch-Godfrey Serial Correlation LM test indicate the presence of significant serial correlation in the model's residuals, as evidenced by the F-statistic (42.49738) and Obs*R-squared (27.60637), both with probability values of 0.0000. This means that past errors influence current errors, violating the assumption of independence in regression analysis. Additionally, the first lag of the residuals (RESID(-1)) is highly significant with a coefficient of 1.3207 and a p-value of 0.0000, further confirming strong autocorrelation. The second lag (RESID(-2)) is also significant, suggesting that past residuals continue to affect present values. Other independent variables, including Inflation Rate, FDI Inflows as a percentage of GDP, and Exchange Rate, do

not significantly impact the residual behavior. The Durbin-Watson statistic (2.2587) suggests that some degree of autocorrelation remains. Given these findings, the null hypothesis of no serial correlation is rejected, implying that the model suffers from autocorrelation. This issue must be addressed using techniques such as adding lagged variables, applying Generalized Least Squares (GLS), or using Newey-West standard errors to ensure more reliable estimations.

4.2.3.4 Linearity Test

This test checks if a linear relationship exists between the dependent variable and the explanatory variables, showing that the model is well specified. The null hypothesis states that a linear relationship exists between variables. Using the Ramsey RESET Test, we accept the null if the probability value is greater than 0.05, and we reject if the probability value is less than 0.05.

Table 4.10: Ramsey RESET Test Result

	Value	**Df**	**Probability**
t-statistic	**9.018560**	**33**	**0.0000**
F-statistic	**81.33443**	**(1, 33)**	**0.0000**

Source: Author's computation using EViews 9

INTERPRETATION

The Ramsey RESET test results indicate that the model suffers from specification error. The t-statistic value of 9.0186 with a p-value of 0.0000 suggests that the squared fitted values significantly impact the dependent variable. Similarly, the F-statistic value of 81.3344 with a p-value of 0.0000 confirms that the model is mis-specified. Since the p-values are less than 0.05, we reject the null hypothesis that the model is correctly specified. This implies that important variables

or nonlinear relationships might be missing from the regression model, suggesting the need for a model re-specification.

4.2.3.5 Heteroscedasticity Test

This test is conducted to check whether the variance of the residuals is unequal or to ensure the residuals have a constant variance. The null hypothesis states that there is no heteroscedasticity (there is homoscedasticity), while the alternative states that there is heteroscedasticity. If the probability value is greater than 0.05, the null hypothesis is accepted.

Table 4.11: Heteroscedasticity Test Result

Statistic	Value	Probability
F-statistic	**9.535134**	**0.0001 (Prob. F(3,34))**
Obs*R-squared	**17.36280**	**0.0006 (Prob. Chi-Square(3))**
Scaled explained SS	**18.75691**	**0.0003 (Prob. Chi-Square(3))**

Source: Author's computation using EViews 9

INTERPRETATION

The Breusch-Pagan-Godfrey test for heteroskedasticity reveals statistically significant results, indicating the presence of heteroskedasticity in the regression model. The F-statistic of 9.5351 has a corresponding probability value of 0.0001, which is well below the conventional significance levels, suggesting that the null hypothesis of homoskedasticity is rejected. Additionally, the Obs*R-squared statistic of 17.3628 and its associated p-value of 0.0006 further confirm this finding. The Scaled Explained Sum of Squares (SS) statistic, which stands at 18.7569 with a probability of 0.0003, also supports the conclusion that variance in the residuals is not constant. These results imply that the model's error terms exhibit heteroskedasticity, which could affect the

efficiency of standard errors and, consequently, the reliability of hypothesis testing. Therefore, corrective measures such as robust standard errors or generalized least squares may be necessary to address this issue.

4.3 Discussion of Findings

4.3.1 Results of Research Questions

This section provides a detailed discussion of the study's findings in relation to the research questions. The empirical results, trend analyses, and statistical tests offer critical insights into the relationships between exchange rate fluctuations, inflation, and foreign direct investment (FDI) inflows in determining Nigeria's Real GDP. The findings are evaluated based on both short-run and long-run effects, considering economic policies, structural challenges, and historical patterns.

4.3.1.1 Research Question 1: How does Exchange Rate fluctuations influence Real GDP in Nigeria?

The findings from the study reveal that exchange rate fluctuations significantly influence Real GDP in Nigeria, but the nature of this impact varies over time. The long-run estimation results indicate that exchange rate depreciation has a positive effect on GDP. This suggests that a weaker naira, by making Nigerian exports cheaper in the global market, may improve trade balances and stimulate economic growth. A depreciating currency can enhance the competitiveness of the non-oil export sector, encourage local production, and attract foreign investments looking for cost advantages.

However, the short-run analysis presents a contrasting view. In the short term, exchange rate fluctuations appear to exert a negative effect on GDP, particularly through lagged values. This implies that while currency depreciation may eventually benefit the economy, its immediate effects tend to create uncertainty and economic instability. Sharp fluctuations in the exchange rate often

result in increased costs of imported goods, higher inflationary pressures, and capital flight, all of which negatively impact economic growth. The findings highlight that excessive exchange rate volatility, rather than the absolute level of the exchange rate, is a critical concern for economic policymakers.

The trend analysis further supports this conclusion. Over the years, Nigeria has experienced multiple phases of exchange rate depreciation, especially following the introduction of the Structural Adjustment Program (SAP) in 1986. The transition from a fixed to a flexible exchange rate system led to rapid depreciation, which persisted into the early 2000s. The stability observed in the mid-2000s was largely due to improved foreign reserves, oil price booms, and proactive monetary policies by the Central Bank of Nigeria (CBN). However, the trend from 2016 onwards shows increasing exchange rate volatility, particularly after the 2020 COVID-19 pandemic and the liberalization of the foreign exchange market.

These findings suggest that while a depreciated exchange rate may enhance economic growth over time, ensuring stability in the foreign exchange market is crucial to minimizing short-term economic disruptions. Policymakers need to adopt a balanced approach that supports a competitive exchange rate while implementing measures to reduce excessive fluctuations, such as building foreign reserves, encouraging non-oil exports, and managing speculative activities in the currency market.

4.3.1.2 Research Question 2: What Is the Impact of Inflation Rate on Real GDP in Nigeria?

The relationship between inflation and Real GDP in Nigeria is found to be negative, but statistically insignificant, in both the short run and the long run. This suggests that while inflation may have an adverse effect on economic growth, its direct impact is relatively weak when

compared to other macroeconomic variables such as exchange rate movements and investment flows.

Theoretically, inflation can influence GDP through multiple channels. Moderate inflation is often seen as a sign of economic expansion, while excessive inflation erodes purchasing power, increases production costs, and discourages investment. In the Nigerian context, the study's results indicate that inflation alone does not have a decisive influence on GDP growth, which suggests that other structural factors may play a more significant role in shaping economic performance.

The trend analysis of inflation over the study period (1986–2023) highlights periods of extreme volatility. In the late 1980s and early 1990s, Nigeria experienced high inflation, peaking at over 70%, primarily due to currency devaluation, fiscal imbalances, and structural adjustment policies. However, the early 2000s saw a period of relative stability, driven by improved monetary policies and fiscal discipline. Between 2005 and 2020, inflation fluctuated moderately but remained within manageable levels, largely influenced by exchange rate movements, global commodity prices, and domestic production levels. From 2020 onwards, inflationary pressures have intensified due to post-COVID-19 economic disruptions, food supply chain challenges, and depreciation of the naira.

These findings suggest that while inflation is a macroeconomic concern, its impact on GDP may be overshadowed by other economic factors, such as investment levels, government spending, and trade policies. The results imply that a singular focus on controlling inflation without addressing the underlying structural weaknesses in the economy may not yield significant improvements in GDP growth. Therefore, inflation management should be integrated with broader economic policies aimed at stabilizing exchange rates, increasing domestic production, and reducing dependence on imported goods.

4.3.1.3 Research Question 3: To What Extent Do Foreign Direct Investment (FDI) Inflows Affect Real GDP in Nigeria?

One of the most striking findings of this study is the significant negative relationship between FDI inflows and Real GDP in both the short and long run. Contrary to conventional economic expectations that FDI boosts economic growth by providing capital, technology, and employment opportunities, the study's results suggest that increased FDI inflows may not necessarily translate into improved GDP growth in Nigeria.

Several factors may explain this paradox. First, while FDI brings in foreign capital, a significant portion of the profits generated by multinational corporations may be repatriated to their home countries rather than being reinvested in the domestic economy. This capital outflow can offset the expected benefits of FDI, limiting its contribution to local economic growth. Second, Nigeria's economic structure may not be adequately positioned to absorb FDI effectively. Weak linkages between foreign firms and domestic industries, policy inconsistencies, and infrastructural deficits may prevent FDI from having a meaningful impact on GDP growth.

The trend analysis of FDI inflows shows a gradual increase, particularly after the liberalization policies of the 1990s and early 2000s. Reforms aimed at attracting foreign investments in sectors such as telecommunications, banking, and oil production contributed to this rise. However, despite the steady increase in FDI, the expected positive impact on economic growth remains limited. Periodic declines in FDI, especially during economic crises and periods of policy uncertainty, further highlight the vulnerability of Nigeria's investment climate.

These findings emphasize the need for policy measures that ensure FDI contributes meaningfully to economic development. This includes improving the ease of doing business, strengthening domestic industries to integrate with foreign investments, and implementing policies that

encourage profit reinvestment within Nigeria. Additionally, attracting FDI into non-oil sectors, such as manufacturing and agriculture, can help diversify the economy and reduce dependency on extractive industries.

4.4 General Discussion

The findings from this study provide critical insights into the relationships between exchange rate fluctuations, inflation, foreign direct investment (FDI) inflows, and Real GDP in Nigeria. The empirical results, supported by trend analyses and statistical tests, reveal complex interactions among these macroeconomic variables, highlighting both short-run and long-run effects. The study's results have significant implications for economic policy and sustainable growth in Nigeria.

Macroeconomic Dynamics and Economic Growth in Nigeria

The results confirm that exchange rate movements play a crucial role in shaping Nigeria's economic performance. In the long run, exchange rate depreciation appears to positively impact Real GDP, suggesting that a weaker naira may enhance export competitiveness and stimulate growth. However, the short-run findings indicate that excessive exchange rate volatility negatively affects GDP, implying that uncertainty in the foreign exchange market may disrupt investment flows, increase inflationary pressures, and create economic instability. The trend analysis further reveals that exchange rate depreciation has been a persistent challenge in Nigeria, particularly since the implementation of the Structural Adjustment Program (SAP) in 1986. In recent years, exchange rate instability has intensified due to external shocks, declining foreign reserves, and policy shifts. These findings emphasize the need for exchange rate policies that strike a balance between maintaining competitiveness and ensuring macroeconomic stability.

Inflation, another key macroeconomic variable, is found to have a negative but statistically insignificant impact on Real GDP. This suggests that while inflation can erode purchasing power and disrupt production, its direct effect on GDP is not as strong as other economic factors. The trend analysis indicates that inflation in Nigeria has been highly volatile, particularly in the 1990s when it peaked at over 70% due to currency devaluation, fiscal deficits, and policy inconsistencies. More recently, inflation has been influenced by exchange rate depreciation, global supply chain disruptions, and rising food prices. Despite monetary policy efforts by the Central Bank of Nigeria (CBN), inflationary pressures remain a significant challenge. The study's findings suggest that while inflation control is important, it should be complemented by broader structural reforms that address supply-side constraints, improve domestic production, and reduce Nigeria's dependence on imported goods.

The most surprising finding of the study is the significant negative relationship between FDI inflows and Real GDP in both the short and long run. Contrary to conventional economic theory, which suggests that foreign investment fosters growth by providing capital, technology, and employment, the results indicate that increased FDI inflows may not necessarily translate into higher GDP growth in Nigeria. Several factors may explain this phenomenon. First, the nature of FDI in Nigeria, particularly in the oil and extractive industries, may not generate substantial linkages with the local economy, limiting its positive spillover effects. Second, a significant portion of the profits generated by multinational corporations may be repatriated to their home countries rather than reinvested in the domestic economy. Third, the weak business environment, policy uncertainties, and infrastructural deficiencies may prevent Nigeria from fully harnessing the potential benefits of foreign investment. The trend analysis confirms that while FDI inflows have generally increased over time, their contribution to economic growth remains questionable. These

findings suggest that Nigeria needs to implement targeted policies that ensure FDI contributes meaningfully to economic development, such as encouraging reinvestment, strengthening local industries, and improving the ease of doing business.

Policy Implications and Economic Recommendations

The findings of this study have significant policy implications. First, exchange rate management should be a priority for policymakers. While currency depreciation can enhance export competitiveness, excessive volatility must be addressed to maintain investor confidence and economic stability. Strengthening foreign reserves, diversifying export earnings, and implementing effective monetary policies can help achieve exchange rate stability.

Second, inflation control should go beyond monetary policies and include structural reforms aimed at increasing domestic production. High inflation in Nigeria is often driven by supply-side constraints rather than excessive demand. Addressing factors such as poor infrastructure, weak agricultural productivity, and dependence on imported goods can help stabilize prices in the long run.

Third, FDI policies should be restructured to ensure that foreign investments contribute more effectively to economic growth. Nigeria needs to attract FDI into productive sectors such as manufacturing, agriculture, and technology, rather than relying solely on investments in the oil and gas sector. Additionally, policies should encourage profit reinvestment within the country and enhance local capacity to benefit from foreign investments.

4.4.1 Discussion with respect to key theory

The findings of this study align with the Optimum Currency Area (OCA) theory, which postulates that countries or regions with closely integrated economies, high labor mobility, and synchronized

economic cycles benefit from a shared currency or fixed exchange rate system. Nigeria, as part of the Economic Community of West African States (ECOWAS), has often considered deeper monetary integration, including a potential common currency, to enhance economic stability and trade. However, this study's results highlight several challenges and implications in the Nigerian context.

A key argument in the OCA theory is that exchange rate stability fosters economic growth by reducing transaction costs and exchange rate uncertainty. The study finds that Nigeria's exchange rate volatility negatively affects economic growth due to inflationary pressures, investment uncertainty, and trade imbalances. This instability suggests that Nigeria, as a single economy, struggles with exchange rate fluctuations, making the idea of a fixed exchange rate or common currency within ECOWAS challenging unless macroeconomic convergence is achieved.

Another core assumption of the OCA theory is labor mobility, which helps absorb asymmetric economic shocks within a currency area. In Nigeria, however, labor mobility across states and regions remains limited due to structural challenges, such as weak infrastructure, regulatory barriers, and cultural differences. This lack of labor flexibility implies that Nigeria may struggle to benefit from a fixed exchange rate system, as internal shocks would not be easily offset by workforce movement.

Trade openness is another essential criterion for an optimum currency area. The study finds that Nigeria's trade is heavily dependent on oil exports, making the economy highly susceptible to external shocks, such as global oil price fluctuations. This exposure weakens Nigeria’s ability to maintain exchange rate stability and raises questions about the feasibility of a fixed exchange rate or monetary union. Unlike highly diversified economies that fit into the OCA framework, Nigeria's dependence on a single commodity makes it more vulnerable to trade imbalances.

The OCA theory also emphasizes the need for fiscal transfers and economic policy coordination to stabilize regions affected by asymmetric shocks. However, Nigeria's fiscal policies have been inconsistent, often driven by short-term political considerations rather than long-term macroeconomic stability. The study reveals that government interventions in exchange rate management, such as the adoption of multiple exchange rate regimes, have exacerbated currency instability rather than provided the stability envisioned under the OCA framework.

Moreover, the assumption that economies within a currency area should experience symmetric economic shocks does not hold for Nigeria. The study highlights that different sectors of the Nigerian economy experience exchange rate fluctuations differently—while the oil sector benefits from currency depreciation, other industries, such as manufacturing and agriculture, face rising costs due to import dependency. This asymmetric response further weakens Nigeria’s suitability for a fixed exchange rate or currency union under the OCA model.

Finally, the study identifies political and institutional challenges as key barriers to exchange rate stability in Nigeria. The OCA theory assumes that participating economies will align their monetary and fiscal policies to achieve macroeconomic stability. However, Nigeria's history of policy inconsistency, lack of monetary policy independence, and corruption undermines its ability to sustain a stable exchange rate regime in line with the OCA framework.

In conclusion, while the OCA theory provides valuable insights into the potential benefits of exchange rate stability for economic growth, Nigeria’s current economic structure, policy environment, and external vulnerabilities present significant barriers to achieving such stability. The findings suggest that rather than adopting a rigid exchange rate system or participating in a monetary union, Nigeria should focus on macroeconomic reforms, diversification, and policy consistency to manage exchange rate fluctuations effectively.